Road Accident Data Analysis & Visualization

An exploratory study of accident patterns, severity and location.

ROAD SAFETY

DATA
↓
INSIGHT

PROJECT TEAM

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2026

Introduction

Understanding the patterns behind reported road accidents

Road accidents are influenced by several interacting factors, including location, time, road conditions and human behaviour. Exploring accident records helps organise these details into patterns that can be examined.

WHY ANALYSE THE DATA?

01 **Identify locations**

Find areas with comparatively high recorded incidents.

02 **Understand timing**

Compare patterns across hours, days and months.

03 **Study outcomes**

Separate accident frequency from injury severity.

Problem Statement

Why accident data needs structured exploration

01	**Fragmented information**	Records may be spread across different sources and formats.
02	**Unclear accident hotspots**	High-frequency locations are difficult to spot without comparison.
03	**Changing risk periods**	Accident patterns may vary by time, weekday or season.
04	**Severity is not captured by totals**	A simple count does not describe the human impact.
05	**Incomplete context**	Weather, road design and traffic exposure may not be available.

Project Objectives

The key questions guiding the analysis

01 **Summarise accident records**

Describe the size and structure of the selected dataset.

02 **Compare accident severity**

Examine fatal, serious and minor outcomes separately.

03 **Explore time-based trends**

Compare incidents by hour, weekday, month or year.

04 **Examine location patterns**

Identify districts, roads or junctions needing attention.

05 **Present useful insights**

Communicate findings through clear charts and observations.

Data & Features

Fields that may be used to describe each accident record

Date and time

Accident date, reporting time or time of day

Location

District, road name, junction or coordinates

Accident type

Collision, overturning or other recorded category

Severity

Fatal, serious, minor or property-damage only

Vehicles

Vehicle types and number involved

Weather

Clear, rain, fog or other recorded conditions

Lighting

Daylight, street-lit or unlit conditions

Road surface

Dry, wet or other recorded road state

Casualties

Number of people injured or killed

The actual fields depend on the dataset selected; unavailable variables should not be assumed.

Accident Pattern Analysis

Dimensions to examine before drawing conclusions

TIME

Hour of day

Compare morning, afternoon, evening and night records.

PLACE

Geographic area

Compare districts, road segments and junctions.

IMPACT

Severity level

Examine fatal, serious and minor outcomes separately.

ADDITIONAL CONTEXT

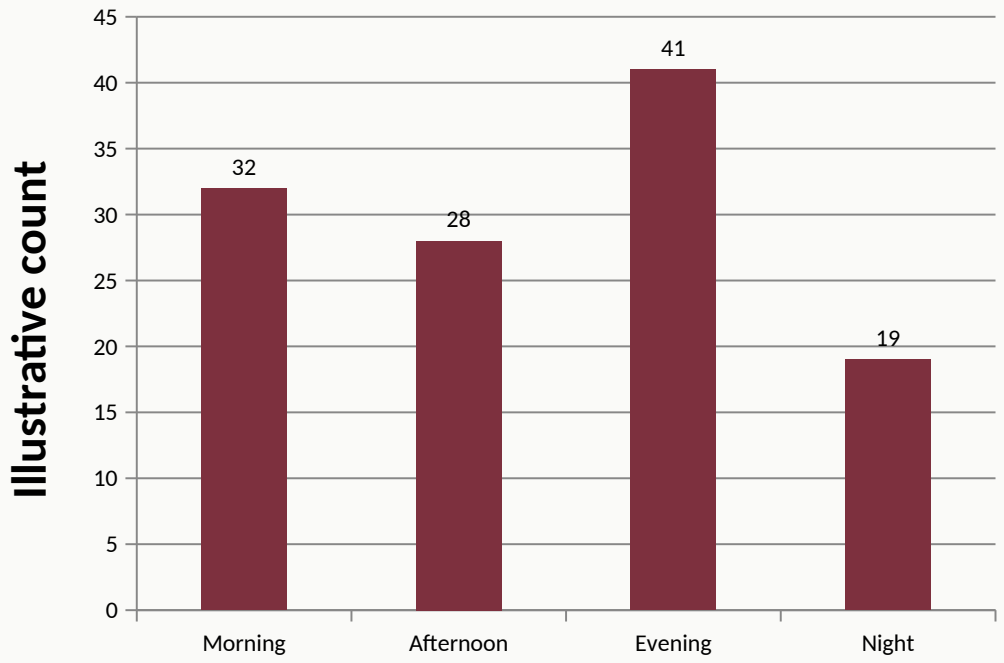
Where recorded, weather, lighting, road surface, vehicle type and traffic volume can help interpret observed patterns.

Data Analysis & Visualization

Use the chart type that best answers each analytical question

Illustrative accident counts by time period

Example values only — not measured dataset results.



RECOMMENDED VISUAL

Bar chart	Compare districts, accident types or severity.
Line chart	Track monthly or yearly changes.
Heatmap	Compare accident frequency by hour and weekday.
Map	Display geographic clusters and potential hotspots.

Key Findings to Investigate

Questions the completed analysis should answer

01	**Where are incidents concentrated?**	Identify locations with higher recorded accident frequency.
02	**When do incidents occur most often?**	Compare hours, weekdays and months using the available records.
03	**Which outcomes are most severe?**	Review fatal, serious and minor categories independently.
04	**What conditions appear alongside incidents?**	Examine weather, lighting and road-surface fields if available.
05	**How reliable are the records?**	Check missing values, duplicate entries and inconsistent labels.

These are analysis questions, not conclusions from a specific dataset.

Recommendations

How evidence can inform practical road-safety measures

01 Review high-frequency locations

Assess signage, road layout, lighting and speed management.

02 Focus on identified risk periods

Align awareness and enforcement with validated time patterns.

03 Promote safe road behaviour

Encourage helmets, seat belts, safe speeds and reduced distraction.

04 Improve accident reporting

Use consistent categories and accurate location information.

05 Evaluate safety interventions

Compare trends before and after changes, considering traffic exposure.

06 Share findings with stakeholders

Communicate evidence to traffic police and local road authorities.

Turning accident records into road-safety insight.

Exploratory data analysis helps describe where, when and how road accidents are reported. Clear visualisations make these patterns easier to interpret and discuss.

DESCRIBE

Summarise accident patterns.

INTERPRET

Identify trends and areas for review.

INFORM

Support evidence-based safety action.

Thank you | Questions

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